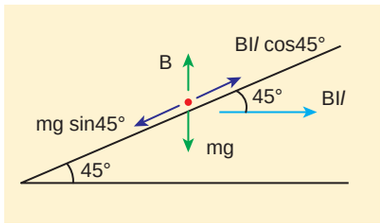
Solution

The linear density of the rod i.e., mass per unit length of the rod is 0.25 kg m^{-1}

$$\Rightarrow \frac{m}{l} = 0.25 \text{ kg m}^{-1}$$

Let I be the current flowing in the metallic rod. The direction of electric current is into the paper. The direction of magnetic force IBl is given by Fleming's left hand rule.



For equilibrium,

$$mg \sin 45^\circ = IBl \cos 45^\circ$$

$$\Rightarrow I = \frac{1}{B}\frac{m}{l} g \tan 45^\circ$$

$$= \frac{0.25 \text{ kg m}^{-1}}{0.25\, T} \times 1 \times 9.8 \; m s^{-2}$$

$$\Rightarrow I = 9.8\, A$$

So, we need to supply current of 9.8 A to keep the metallic rod stationary.

3.10.6 Force between two long parallel current carrying conductors

Two long straight parallel current carrying conductors separated by a distance r are kept in air as shown in Figure 3.58. Let I_1 and I_2 be the electric currents passing through the conductors A and B in same direction (i.e. along z - direction) respectively. The net magnetic field at a distance r due to current I_1 in conductor A is

$$\vec{B}_1 = \frac{\mu_\circ I_1}{2\pi r}(-\hat{i}) = -\frac{\mu_\circ I_1}{2\pi r}\hat{i}$$

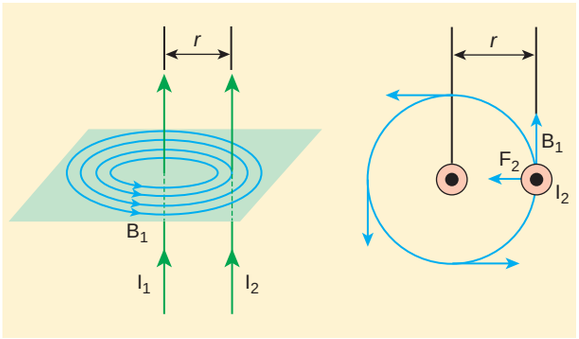


Figure 3.58 Two long straight parallel wires

From thumb rule, the direction of magnetic field is perpendicular to the plane of the paper and inwards (arrow into the page ⊗) i.e. along negative $\hat{i}$ direction.

Let us consider a small elemental length dl in conductor B at which the magnetic field $\vec{B}_1$ is present. From equation 3.65, Lorentz force on the element dl of conductor B is

$$d\vec{F} = \left(I_2 d\vec{l} \times \vec{B}_1\right) = -I_2 dl \frac{\mu_\circ I_1}{2\pi r}(\hat{k} \times \hat{i})$$

$$= -\frac{\mu_\circ I_1 I_2 dl}{2\pi r}\hat{j}$$

Therefore the force on dl of the wire B is directed towards the wire W_1. So the length dl is attracted towards the conductor A. The force per unit length of the conductor B due to the wire conductor A is

$$\frac{\vec{F}}{l} = -\frac{\mu_\circ I_1 I_2}{2\pi r}\hat{j}$$

In the same manner, we compute the magnitude of net magnetic induction due to current I_2 (in conductor A) at a distance r in the elemental length dl of conductor A is

$$\vec{B}_2 = \frac{\mu_\circ I_2}{2\pi r}\hat{i}$$

From the thumb rule, direction of magnetic field is perpendicular to the plane of the paper and outwards (arrow out of the page ⊙) i.e., along positive $\hat{i}$ direction.

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